When Structure Hurts: Understanding Evidence Loss in Retrieval-Augmented Reasoning

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Abstract

Retrieval-augmented systems increasingly transform retrieved passages into claims, reasoning paths and graphs before generation, but such transformations can also lose facts or add misleading relations. Because system-level comparisons change retrieval, content and formatting at once, the effect of the representation itself is rarely measured.

We present the same gold supporting evidence for multi-hop questions in seven representations, under an information-controlled protocol that verifies identical facts per question and a budget-controlled protocol that records which facts each format can afford. We complement this with retrieved-evidence baselines and controlled perturbations. After an exploratory pilot, we tested six preregistered contrasts on a disjoint sample of 4,976 MuSiQue and 2WikiMultiHopQA questions with Qwen2.5-7B-Instruct.

Neither graph-derived ordering nor explicit derived relations changed exact match detectably; on 2WikiMultiHopQA both effects are equivalent to zero within two points. Splitting passages into sentence items lowered 2WikiMultiHopQA exact match by 1.7 points (95% CI $[-2.9, -0.5]$), almost entirely through more frequent abstention. Structure hurt when it was noisy: predicted relations lowered exact match by 3.6 and 4.5 points, a loss concentrated in prompts where the predictor marked compatible facts as contradictory. Under a token budget, structured formats kept 15–20 percentage points fewer facts than passages.

These results suggest that the value of structured evidence lies less in its relations than in what a pipeline keeps or loses, and that the reliability of relation extraction should be measured as carefully as its benefits.

1 Introduction

Retrieval-augmented generation (RAG) gives a language model access to text that its parameters do not contain, by retrieving passages and placing them in the prompt (Lewis et al., 2020). For questions that require combining several facts, as in multi-hop benchmarks such as MuSiQue, 2WikiMultiHopQA and HotpotQA (Trivedi et al., 2022; Ho et al., 2020; Yang et al., 2018), the passages are typically handed to the model as an unordered list and left for the model to link.

A growing family of methods therefore transforms retrieved evidence before generation. Passages are decomposed into propositions or atomic claims (Chen et al., 2024), organised into trees of summaries (Sarthi et al., 2024), or linked into entity and fact graphs whose nodes, relations and paths are serialised into the prompt (Edge et al., 2024; Gutiérrez et al., 2024, 2025; Zhu et al., 2025; Guo et al., 2024; Chen et al., 2026).

Every such transformation can also go wrong. An extractor may omit the sentence that names the answer, and a token budget may cut it because the structured format spends tokens on identifiers and relation lines. Splitting a passage into sentences turns "It stars Anne Hathaway …" into an item whose subject must be recovered from elsewhere. A relation predictor may add a statement that the source never made: in our experiments, it marked "The teaser of the film was released in February 2016" and "The trailer of the film was released in October 2016" as contradicting each other, and once shown that line, the model declared the evidence insufficient for a question it otherwise answered correctly.

Benefits and costs therefore arrive together, and the system-level comparisons in the literature cannot separate them. When a graph-based system outperforms or underperforms passage RAG (Xiang et al., 2026; Han et al., 2025; Zhou et al.,

2025), its retrieval, the content that reaches the model and the way that content is formatted all differ at once.

We argue that the effect of an evidence representation should be measured with the content held fixed, and that the cost of a transformation, measured as the evidence it loses or the information it adds, should be reported separately from its benefit. Therefore, we investigate how the representation of the *same* supporting evidence affects multi-hop answer accuracy, and when transformation errors outweigh the benefits of explicit structure.

To do so, we present the gold supporting evidence of each question in seven representations: passages, sentence items, shuffled items, reasoning steps, graph-ordered items, graph-ordered items with explicit relation lines, and the same with predicted relations. An *information-controlled* protocol verifies, per question, that every representation shows exactly the same facts; a *budget-controlled* protocol gives every representation the same token budget and the same representation-independent ranking of facts, and records which facts each representation can afford. We complement these with retrieved-evidence baselines and controlled perturbations. All design decisions, primary contrasts and statistics were fixed before generation. After an exploratory pilot, we tested the six primary contrasts on a disjoint confirmatory sample of 4,976 questions, with sample sizes set from the pilot variance.

Contributions. (i) A controlled framework that separates the information a representation carries from its formatting and from the information its structure adds, with per-question verification of fact identity and accounting of evidence loss under a budget. (ii) A pre-registered test with mostly null results: for Qwen2.5-7B-Instruct, neither graph-derived ordering nor explicit derived relations changed exact match detectably on MuSiQue or 2WikiMultiHopQA, while splitting passages into sentence items lowered 2WikiMultiHopQA exact match by 1.7 points, almost entirely through increased abstention. (iii) Evidence that noisy structure hurts: predicted relations lowered exact match by four to five points, a loss traced to spurious contradiction relations, and under a token budget structured formats kept 15–20 percentage points fewer facts than passages. (iv) A release of code, frozen question sets, run manifests and analysis outputs, so that every number can be traced to a run.

2 Related Work

Passages, finer units and compression. Retrieval-augmented generation conditions a language model on retrieved text (Lewis et al., 2020), usually by concatenating passages in rank order. This presentation is not neutral: models use information in the middle of a long context less reliably than at its edges (Liu et al., 2024), and irrelevant retrieved passages change generation quality in ways that retrieval scores do not predict (Cuconasu et al., 2024). Dense X Retrieval indexes propositions, i.e. atomic, self-contained factoids, and finds that finer units improve retrieval and downstream QA under a fixed computation budget (Chen et al., 2024); FActScore uses a similar decomposition for evaluation (Min et al., 2023). Compression methods trade information for length explicitly: LongLLMLingua prunes prompts conditioned on the question (Jiang et al., 2024), and ECoRAG compresses retrieved documents by evidentiality and retrieves more when the result is insufficient (Jeong et al., 2025). Our Protocol B shares this view of a budget, but holds the selection policy fixed across representations, so that the cost of formatting is separated from the choice of content.

Graph-based and hierarchical retrieval. GraphRAG builds an entity graph with community summaries (Edge et al., 2024), RAPTOR a tree of recursive summaries (Sarthi et al., 2024), and HippoRAG and HippoRAG 2 a knowledge graph traversed with personalised PageRank (Gutiérrez et al., 2024, 2025). KG^{2}RAG links chunks through fact-level relations (Zhu et al., 2025), LightRAG combines graph indexing with dual-level retrieval (Guo et al., 2024), PathRAG retrieves, prunes and serialises relational paths (Chen et al., 2026), and LinearRAG avoids relation extraction, arguing that extracted relations are unreliable and costly (Zhuang et al., 2026). In these systems graph construction, retrieval and prompt formatting change together.

When do graphs help? GraphRAG-Bench is motivated by reports that GraphRAG often underperforms vanilla RAG on real-world tasks. It evaluates the whole pipeline, from graph construction to generation, and identifies the scenarios in which graph structures give measurable benefits (Xiang

et al., 2026). A benchmark comparison of RAG and GraphRAG under a unified protocol finds that each has distinct strengths depending on the task (Han et al., 2025), and a unified framework compares graph-based RAG methods under the same experimental settings (Zhou et al., 2025).

These studies compare *systems*: when a graph method wins or loses, retrieval, the content that reaches the model and its serialisation all differ at once. Our study is narrower and complementary. We fix the source evidence, verify that every representation shows the same facts, and vary only the representation, or, under a budget, only how much of a fixed ranking each representation can afford. This isolates presentation and serialisation overhead, at the price of not measuring retrieval benefits.

Datasets, attribution and our earlier architecture. We evaluate on MuSiQue (Trivedi et al., 2022) and 2WikiMultiHopQA (Ho et al., 2020), with HotpotQA (Yang et al., 2018), HoVer (Jiang et al., 2020) and FEVER (Thorne et al., 2018) as secondary datasets. ALCE evaluates whether generated statements are supported by cited passages (Gao et al., 2023); we record citations but do not analyse them. Experiment 0 includes a reconstruction of EG-RAG, an evidence-graph pipeline, only as a retrieved-evidence baseline. Its unspecified components are our own choices (Appendix C), and no result of the original architecture paper is reused.

3 Problem Formulation

Structured evidence pipelines usually change two things at once: *which facts* reach the answerer, and *how* those facts are arranged and annotated. A transformation can lose a fact (an extractor omits it, a budget cuts it), and it can add information that is not a fact of the source text (an order, a grouping into steps, a relation between two facts). We measure the two separately.

Facts and representations. For a question q with gold answers A_q we start from a fixed set of source passages; in our controlled experiments these are the dataset's gold supporting passages. Segmenting them into sentences gives the fact set $F_q = \{f_1, \ldots, f_n\}$, where each fact keeps the identifier of its source span. Because facts are sentences of the gold passages, not model-generated paraphrases, F_q contains no extraction error; model-based extraction enters only through the retrieved-evidence systems of Experiment 0. A representation ρ maps the facts, optionally with structure S_q derived from annotations or predicted by a model, to an evidence block

$$E_q^\rho = \mathrm{render}_\rho\big(\pi_\rho(F_q),\, S_q\big), \tag{1}$$

where π_ρ orders and groups the facts and render_ρ serialises them, possibly with a list of relations. We write $\mathrm{facts}(E)$ for the facts that appear in a block, and $\mathrm{added}(\rho)$ for the information that the block carries beyond them: an order, a grouping into steps, or relation lines. A fixed answerer M with one prompt template τ, identical for every representation, produces $\hat{a} = M(\tau(q, E_q^\rho))$ with greedy decoding.

Protocol A: information-controlled. Protocol A compares representations whose fact sets are identical, $\mathrm{facts}(E_q^\rho) = F_q$ for every ρ, which we verify per question by comparing identifiers and text digests. Under this constraint an accuracy difference cannot come from a missing fact; it can still come from $\mathrm{added}(\rho)$, which we record per record, and from formatting tokens. We record the content tokens $c_q = \sum_j |\mathrm{text}_j|$ and the overhead $o_q^\rho = |E_q^\rho| - c_q$ separately.

Protocol B: budget-controlled. Under a budget $B_q = \beta\, c_q$ we rank the facts once, with BM25 against the question, independently of the representation and without gold labels, and each condition keeps the longest prefix of the ranking whose rendering fits,

$$k_\rho = \max\{k :\ |\mathrm{render}_\rho(\pi_\rho(F_q^{\le k}), S_q)| \le B_q\}. \tag{2}$$

Under the *per-representation* policy each representation keeps its own k_ρ, so verbose formats keep fewer facts; under the *shared* policy all keep $\min_\rho k_\rho$ and show the same facts. Facts are never cut mid-sentence. For every dropped fact we record whether it is a gold supporting fact and whether it contains the answer string; this is our measure of information loss.

Perturbation validity and objective. An intervention δ maps an evidence block E to a corrupted block $\delta(E)$. It is *label-preserving* if A_q remains correct given $\delta(E)$. Reordering facts or relabelling a relation is label-preserving; deleting a required fact is not. Accuracy changes are estimated only on label-preserving records that do not introduce

the answer through a distractor. For a contrast between ρ and ρ' we estimate the paired difference in question-level exact match over the questions Q available in both conditions, with unanswered questions scored as incorrect:

$$\Delta_{\rho'-\rho} = \frac{1}{|Q|} \sum_{q \in Q} \big(\mathrm{EM}(q, \rho') - \mathrm{EM}(q, \rho)\big). \quad (3)$$

4 Experimental Framework

Figure 1 shows the pipeline in the order in which a question passes through it.

Questions and structure. Questions come from frozen samples of the development splits, drawn with a fixed seed before any generation and stored with the hashes of the dataset files; the pipeline refuses to run on any question outside a frozen file or on the questions used during prompt development. The MuSiQue decomposition and the 2WikiMultiHopQA evidence triples give an order of reasoning steps, from which we *derive* dependencies between facts by rule. Where a dependency can be anchored to a fact by an exact string match we call it *exact-anchor derived*, otherwise *fallback-anchor derived*. No edge in this study is human-verified.

Retrieval and extraction (Experiment 0). In the distractor setting each question has its own passage pool (20 passages for MuSiQue, 10 for 2WikiMultiHopQA). B0 keeps the top four BM25 passages (Robertson and Zaragoza, 2009); B1 reranks the top ten with a cross-encoder and keeps four. B2 rewrites every sentence of the B1 passages into short facts with the same generator; B3 orders those facts along a predicted reasoning path; B4 is our reconstruction of the EG-RAG evidence-graph pipeline, which retrieves with a hybrid sparse–dense score, extracts claims, predicts typed relations, propagates belief and selects a token-budgeted subgraph (Appendix C).

Representations (Experiments 1 and 2). Table 1 defines the representations; Appendix B shows examples. R5 orders the sentence items by a topological order of the derived dependency graph at passage level, and R4 adds a `Relations:` block listing the derived dependencies. R5−R1 therefore measures the effect of the graph-derived order without relation lines, and R4−R5 the effect of explicit relations at a fixed order. R4P renders the same facts with *predicted* relations: typed relations (support, contradiction, duplication, supersession) from an NLI cross-encoder over candidate pairs, plus a title-mention dependency heuristic, with the order recomputed from the predicted graph.

ID	Rendering of the same fact set
R0	one item per gold passage, dataset order
R1	one item per sentence, dataset order
R2	R1 shuffled (seeds 0, 1, 2; averaged per question)
R3	R1 grouped under `Step` k`:` in reasoning order
R5	R1 in graph-derived (topological) order
R4	R5 plus a `Relations:` block of derived dependencies
R4P	R4 with model-predicted relations and order

Table 1: Representations. Every item carries a per-prompt identifier (`[E1]`, ...) and its source title.

Selection, generation and evaluation. Under Protocol A every representation shows the complete fact set; under Protocol B the budget is $\beta \in \{1.0, 0.5\}$ times the content tokens. A budget of $1.0\times$ the content tokens is already smaller than the full rendering of any representation that spends tokens on identifiers, titles or relation lines. The prompt (Appendix A) is identical for every representation, asks for an answer and the identifiers of the items used, and allows the answer *insufficient evidence*. A guard checks for every record that the prompt equals the template filled with the evidence block and the question, so no gold answer can enter the prompt through another channel. Exact match (EM) after SQuAD normalisation (Rajpurkar et al., 2016), maximised over aliases, is the primary metric for QA, and verdict accuracy for HoVer and FEVER. Secondary metrics are token F1, coverage (the fraction of questions with a parsed, non-abstaining answer), answer-in-evidence, and evidence and overhead tokens. Every attempted question stays in the denominator.

Perturbations (Experiment 2). Interventions applied to the R1 and R4 renderings of the gold evidence remove required facts (P1), retype or flip relations (P2), permute positions (P3), insert non-gold sentences from the question's own pool as distractors (P4), strip source titles (P5), or delete a negation or replace an entity with a pronoun (P6). Temporal-conflict interventions (P7) require verified temporal data and are unavailable. The component-attribution experiment (Experiment 3)

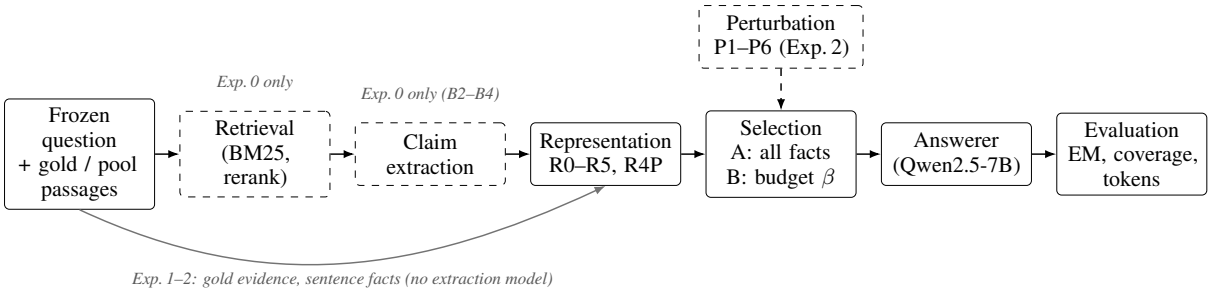


Figure 1: The evidence pipeline. Experiment 1 starts from the gold passages and varies only the representation (Protocol A), or the representation under a fixed fact ranking and token budget (Protocol B). Experiment 0 runs the retrieved-evidence pipeline for five systems; Experiment 2 corrupts the evidence of two representations. Dashed boxes are used only by some conditions.

requires human-verified structure and was not executed, and we did not evaluate an adaptive representation selector (see Limitations).

5 Experimental Setup

Questions. The pilot sample (seed 1), fixed before any generation, contains 100 MuSiQue questions stratified by hop count, 100 2WikiMultiHopQA questions stratified by question type and, as secondary datasets, 50 questions each from HotpotQA, HoVer and FEVER; questions without gold evidence or (for QA) a reasoning order are ineligible, and the fifteen questions used during prompt development are excluded. We first run the 200 MuSiQue and 2WikiMultiHopQA questions (the *focused pilot*) and then the complete 350-question set (the *expanded pilot*); generations shared between the two are served from a cache keyed on the full prompt and every decoding setting, so they are identical by construction. After the pilot we wrote and committed a confirmatory protocol, and only then drew a *confirmatory sample* with a different seed (2) from the questions that the pilot and prompt development had not used: 1,744 MuSiQue questions (hop 2/3/4: 687/686/371, the last stratum exhausting the eligible pool) and 3,232 2WikiMultiHopQA questions (808 per type). Its size is the number of questions needed for 80% power to detect a difference of 0.03 exact match at $\alpha = 0.05/6$, computed from the pilot standard deviation of paired differences inflated by 20%, maximised over the three contrasts of each dataset. The confirmatory run uses Protocol A only.

Answerer. The answerer is Qwen2.5-7B-Instruct (Qwen Team, 2024) at the pinned revision `a09a3545`, in bfloat16, with greedy decoding, at most 64 new tokens and at most 4,096 input tokens. Auxiliary models are bge-small-en-v1.5 (dense retrieval in B4), ms-marco-MiniLM-L6-v2 (reranking), nli-deberta-v3-xsmall (relation prediction) and nli-deberta-v3-small (grounding), all pinned (He et al., 2023). Experiment 0 also uses the 7B model as its claim extractor.

Hardware and batch size. All generations run on a single NVIDIA RTX 5000 Ada GPU (32 GB; CUDA 13.0, PyTorch 2.12.1, Transformers 5.12.1) with batch size 1. In a preflight on questions excluded from all reported runs, batched bfloat16 greedy decoding changed 3 of 8 outputs of the 7B model, including one answer, whereas unbatched outputs were identical across runs. Batching would thus have made outputs depend on which prompts share a batch, and batch composition depends on prompt length and hence on the representation.

Statistics. The unit of analysis is the question. For each contrast we report the paired difference in question-level EM over the questions available in both conditions, with a 95% percentile bootstrap interval (2,000 resamples, seed 0) and a two-sided bootstrap p-value (Efron and Tibshirani, 1993); R2 seeds are averaged within a question before pairing. Every contrast also reports the number of questions whose two evidence blocks are byte-identical (for example, R5 equals R1 when the graph-derived order coincides with the dataset order), the same estimate restricted to the remaining *informative* pairs, and the paired difference in coverage. Protocols, datasets, budgets, policies and edge strata are never pooled. As secondary analyses we report the exact McNemar test on discordant pairs (McNemar, 1947) and the standardised paired effect d_z. All these procedures, the pri-

mary metric and the primary contrasts were fixed in a pilot protocol before any generation. The confirmatory family consists of the six primary tests (three contrasts on two datasets) on the confirmatory sample, with Holm correction (Holm, 1979) at a family-wise α of 0.05; we also report whether the 90% interval lies inside a pre-specified equivalence margin of ± 0.02. Everything else, including all pilot results, the secondary contrasts on the confirmatory sample, and the analyses that we added after seeing the confirmatory result, is exploratory and reported as such. An interval that contains zero is reported as inconclusive, never as evidence of equivalence.

6 Results

We report the confirmatory test first, because it is the only part of the study designed to support a claim, and then the exploratory pilot results that contextualise it. Unless stated otherwise, intervals are paired 95% bootstrap intervals over questions, and differences are stated as the first-named representation minus the second-named one.

6.1 Confirmatory test of the primary contrasts

Table 2 reports the six pre-specified tests on the disjoint confirmatory sample of 1,744 MuSiQue and 3,232 2WikiMultiHopQA questions. All 44,784 records were generated without errors, with information equivalence verified for every question.

One of the six tests rejects the null hypothesis after Holm correction. On 2WikiMultiHopQA, presenting the gold passages as sentence items (R1) instead of passages (R0) lowers exact match by 0.017 (95% CI $[-0.029, -0.005]$, $p_{\text{Holm}} = 0.018$); the paired coverage difference is -0.052. On MuSiQue the same contrast is -0.001 $[-0.014, +0.014]$, and its 90% interval lies inside the pre-specified equivalence margin of ± 0.02.

Neither structural contrast is significant on either dataset. Graph-derived ordering (R5−R1) gives -0.011 $[-0.026, +0.003]$ on MuSiQue and -0.008 $[-0.018, +0.002]$ on 2WikiMultiHopQA, and explicit relation lines at a fixed order (R4−R5) give $+0.010$ $[-0.003, +0.024]$ and $+0.006$ $[-0.002, +0.015]$. On 2WikiMultiHopQA both structural contrasts are equivalent to zero within ± 0.02. On MuSiQue neither interval excludes zero, but neither lies inside the margin either, so we can only call them inconclusive.

Two features of the data qualify the structural contrasts. First, many pairs are byte-identical: R5 equals R1 for 1,858 of 3,232 2WikiMultiHopQA questions and 522 of 1,744 MuSiQue questions, and R4 equals R5 for the 879 2WikiMultiHopQA questions without derived edges. Restricted to the pairs whose evidence differs, the estimates are -0.019 (2Wiki, R5−R1), -0.016 (MuSiQue, R5−R1) and $+0.008$ (2Wiki, R4−R5). Second, all structure is derived from dataset annotations, not verified by humans, so a null result concerns derived structure of this quality.

6.2 Where the confirmed effect comes from

Table 3 splits each paired difference according to which of the two conditions produced an answer. This decomposition was added after we saw the confirmatory result and is exploratory. For R1−R0 on 2WikiMultiHopQA, the questions that both representations answered contribute $+0.001$; the effect comes almost entirely from questions answered under only one representation, where the model abstained under the sentence format on 250 questions that it answered under passages, against 81 in the other direction. In other words, for this generator the sentence format did not make answered questions less accurate. It made the model more likely to reply *insufficient evidence* when given exactly the same facts. Abstention is pervasive in general: with complete gold evidence, 710 of 1,744 MuSiQue questions and 529 of 3,232 2WikiMultiHopQA questions are answered under neither R0 nor R1.

6.3 Secondary contrasts on the confirmatory sample

The confirmatory run also generated the secondary representations, whose contrasts were prespecified as secondary and are therefore descriptive (Figure 2; Table 4 in Appendix F).

Replacing derived relations with model-predicted relations (R4P−R4) lowers exact match on both datasets: by 0.036 $[-0.045, -0.027]$ on 2WikiMultiHopQA and 0.045 $[-0.060, -0.030]$ on MuSiQue, with coverage differences of -0.061 and -0.089. The effect of shuffling the sentence items (R2−R1) depends on the dataset: -0.024 $[-0.036, -0.012]$ on 2WikiMultiHopQA but $+0.013$ $[-0.001, +0.028]$ on MuSiQue. Grouping items under reasoning steps (R3−R5) makes no detectable difference (-0.004 and -0.007).

Data	Contrast	n	ΔEM	95% CI	p	p_{Holm}	+/−	Ident.	Inform. Δ	ΔCov.	Equiv.
MuSiQue	R1−R0	1744	-0.001	[-0.014, +0.014]	0.974	0.974	70/71	3	-0.001	+0.009	yes
MuSiQue	R5−R1	1744	-0.011	[-0.026, +0.003]	0.122	0.535	68/88	522	-0.016	+0.003	no
MuSiQue	R4−R5	1744	+0.010	[-0.003, +0.024]	0.117	0.535	75/57	0	+0.010	+0.004	no
2Wiki	R1−R0	3232	-0.017	[-0.029, -0.005]	0.003	0.018	185/239	44	-0.017	-0.052	no
2Wiki	R5−R1	3232	-0.008	[-0.018, +0.002]	0.107	0.535	126/152	1858	-0.019	-0.012	yes
2Wiki	R4−R5	3232	+0.006	[-0.002, +0.015]	0.174	0.535	103/84	879	+0.008	-0.011	yes

Table 2: Confirmatory test of the six pre-specified primary contrasts (Protocol A, EM, disjoint sample; `docs/CONFIRMATORY_PROTOCOL.md`). p: two-sided paired bootstrap; p_{Holm}: Holm-adjusted over the six tests. Equiv.: 90% CI inside ± 0.02 (pre-specified secondary check). Source: `F_exp1_contrasts.csv`.

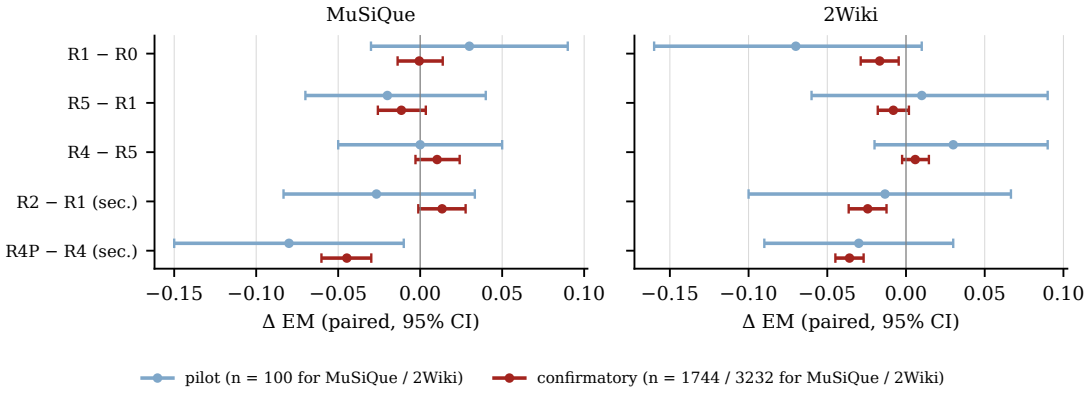


Figure 2: Protocol A contrasts in the exploratory pilot (n=100 per dataset) and on the disjoint confirmatory sample. The pilot estimate of R1−R0 on 2WikiMultiHopQA (-0.07) overstated the confirmed effect (-0.017). Contrasts marked *sec.* are secondary.

Data	Contrast	ΔEM	both	one abst.	abst. 1st/2nd
2Wiki	R1−R0	-0.017	+0.001	-0.018	250/81
2Wiki	R4P−R4	-0.036	-0.011	-0.025	257/59
MuSiQue	R1−R0	-0.001	-0.002	+0.001	94/110
MuSiQue	R4P−R4	-0.045	-0.005	-0.040	218/63

Table 3: Exploratory decomposition (added after the confirmatory result) of paired EM differences on the confirmatory sample into the contribution of questions answered under both representations (*both*) and under only one (*one abst.*); the last column counts questions where only the first- / only the second-named representation abstained. Full table: Appendix F.

6.4 Exploratory pilot results

Protocol B, the retrieved-evidence baselines and the perturbations were run only on the exploratory pilot ($n = 100$ per primary dataset). Appendix E gives the details and Appendix F the tables. *Budget (Protocol B).* Under a budget equal to the content tokens of the fact set, with each representation keeping as much of the shared ranking as it can afford, R0 keeps 0.83 (2WikiMultiHopQA) and 0.81 (MuSiQue) of the facts, against 0.62–0.65 for R1, R4 and R5, whose identifiers, titles and relation lines cost 2.5 to 4 times as many overhead tokens. The answer string consequently remains in 0.85 of R0 blocks on 2WikiMultiHopQA but only 0.64–0.66 of the others; the accuracy differences are smaller than these losses and inconclusive at this n. *Retrieved evidence (Experiment 0).* Extracted claims and predicted paths are indistinguishable from reranked passages in exact match, at about three times the tokens. Our EG-RAG reconstruction, at a similar token count to reranked passages, is less accurate (-0.15 on 2WikiMultiHopQA and -0.09 on MuSiQue), because its budgeted selection keeps the answer string less often and it abstains more. *Perturbations (Experiment 2).* Removing required facts mainly raises abstention, hard distractors lower exact match in every QA cell (by up to 0.16), and retyping 40% of the derived relations lowers 2WikiMultiHopQA exact match by 0.10 $[-0.18, -0.01]$. Reordering and stripping titles have small effects of inconsistent sign.

7 Analysis and Discussion

Structure that adds no facts did not change accuracy detectably. With the facts held fixed, neither the graph-derived order nor the explicit de-

pendency lines changed exact match by more than about one point, and on 2WikiMultiHopQA both effects are equivalent to zero within two points. This observation is narrower than "structure does not help". The structure we test is derived from annotations and therefore mostly correct, and it accompanies facts that already suffice. Structure may matter more for scattered retrieved evidence or for weaker readers, which we did not test. For a 7B instruction-tuned reader on two multi-hop datasets, however, the benefit attributed to graph-structured evidence in system-level comparisons is unlikely to come from the relation lines or the ordering themselves when the same facts are already present.

When structure hurts: predicted relations introduce misleading statements. Predicted relations were the clearest case of a transformation that hurts, and inspection points to a specific mechanism. The NLI-based predictor labels many pairs of compatible gold facts as contradictory: there are 9,688 `CONTRADICTS` lines across the 3,232 2WikiMultiHopQA prompts, and 63% of the prompts contain at least one. For a question asking whether two films come from the same country, it marks "The teaser of the film was released in February 2016" and "The trailer of the film was released in October 2016" as contradicting each other; with this block the model replies *insufficient evidence*, with the derived block it answers correctly. Two successive Zhou kings described as the "nineteenth" and "twentieth" king are likewise marked as contradictory, and the model again abstains. Comparing the same questions under derived and predicted relations (Table 6, Appendix F), the harm is concentrated in prompts that contain a contradiction line. There, abstention rises from 0.32 to 0.42 on 2WikiMultiHopQA and from 0.50 to 0.64 on MuSiQue; elsewhere the two conditions are practically indistinguishable.

Because the two subsets also differ in the number of relation lines, we read this as evidence that *misleading* relation lines drive the effect, not as a separate estimate of a contradiction effect. The pilot perturbations point the same way: deliberately retyping 40% of the derived relations lowered 2WikiMultiHopQA exact match by about ten points. A reader takes explicit relations at face value, so the quality of relation extraction is itself an accuracy risk.

Representation effects run through abstention. Both clear effects in this study came mostly from abstention, not from less accurate answers: the confirmed sentence-item effect and the predicted-relation effect. In 7 of the 12 pilot questions where passages beat sentence items on 2WikiMultiHopQA, the sentence format led the model to declare the evidence insufficient; these questions need two facts from different passages, which the passage format presents as two coherent paragraphs, and splitting can make an item context-dependent ("It stars Anne Hathaway …", Appendix B). For instruction-tuned readers, a large part of what a representation changes is the model's willingness to answer. Accuracy alone hides this, and reporting coverage next to it makes the effect visible.

Serialisation overhead costs evidence under a budget. Under a budget, the formats that spend tokens on identifiers, titles and relation lines kept 15–20 percentage points fewer facts than passages, and dropped answer-bearing facts 1.5 to 3 times as often. Its effect on accuracy was smaller than the evidence loss, because many questions fail even with complete evidence. Budgeted comparisons should therefore report what was shown, not only how accurately the model answered. The same holds for retrieved evidence: our EG-RAG reconstruction lost accuracy at its budgeted selection step, where the answer string survived less often than in reranked passages of similar length, whereas claim extraction itself preserved it.

8 Conclusion

For one 7B instruction-tuned reader on MuSiQue and 2WikiMultiHopQA, a pre-registered test on 4,976 held-out questions found that, with the facts held fixed, graph-derived ordering and explicit derived relations did not change exact match detectably, while sentence items lowered 2WikiMultiHopQA exact match by 1.7 points through abstention. Structure hurt when it was noisy: predicted relations cost four to five points, concentrated where the predictor asserted contradictions between compatible facts. Replicating this with other generators, with human-verified relations, and with retrieved rather than gold evidence are the most useful next steps.

Limitations

One generator. All results come from Qwen2.5-7B-Instruct with greedy decoding and one prompt template. The abstention behaviour that carries most of our effects is a property of this model and this prompt, which explicitly allows the answer *insufficient evidence*. Larger models, other families, or a prompt without an abstention option may respond differently to the same representations. The confirmatory sample is disjoint from the pilot, but it is not an independent replication with a different generator, dataset or team.

Derived, not verified, structure. Every dependency edge was derived from dataset annotations by rule. For a substantial fraction of questions the claim-level endpoint is a fallback guess; this is the stratum C of our reports, and we report contrasts per stratum, but we never treat it as gold. A 50-question human verification package was prepared but not completed, so no result concerns human-verified structure. For the same reason we did not run the component-attribution experiment (Experiment 3), which requires verified structure to be meaningful, and we report no relation precision or recall as a measure of relation quality.

Gold evidence. The controlled experiments start from the gold supporting passages. This isolates representation from retrieval, but it means our null results for structure concern the case in which the facts are complete. Experiment 0 covers retrieved evidence only at pilot scale ($n = 100$ per dataset).

Pilot-scale secondary results. Protocol B, the perturbation experiment, the retrieval baselines and the secondary datasets were run only on the exploratory pilot. Many of their comparisons are underpowered, and they involve many cells, so individual intervals that exclude zero should not be over-interpreted. The pilot overestimated the one effect that we later confirmed by a factor of four.

Metric. Exact match is strict. A manual check found near-misses such as missing diacritics and partial titles, and folding diacritics as a secondary check changed few records and no conclusion. Coverage and token F1 are reported alongside it. The NLI-based groundedness metric gave very low values even on unperturbed evidence, and we do not interpret it.

Analyses added after the confirmatory result. The abstention decomposition and the analysis of predicted contradiction lines were designed after we saw the confirmatory result. They explain the mechanism of effects that were measured independently, but they are exploratory.

Unexecuted experiments. Temporal-conflict perturbations (P7) need verified temporal data, which do not exist for these datasets. Experiment 3 needs verified structure. An adaptive representation selector (Experiment 4) is not evaluated: our results show no representation property that predicts a consistent accuracy gain, so there was nothing for a selector to exploit.

Compute and hardware. All generations ran on a single 32 GB GPU with batch size 1, using about 10 GPU-hours in total. We did not explore sampling-based decoding or multiple seeds of the generator.

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A Prompts

Every Experiment 0–2 answer uses the same prompt (version `qa-v2`); only the evidence block differs between representations. The system message is *You answer using only the evidence you are given.* The user message is:

```
Use only the evidence below to answer the question.
Each evidence item has an ID such as [E1]. Items may
be grouped into steps or followed by a list of
relations between items.

Evidence:
{evidence}

Question: {question}

Reply in exactly two lines:
Answer: <short answer: a name, date, number, short
phrase, or yes/no>
Evidence: <IDs of the items you used, e.g. E1, E2>
If the evidence is not sufficient, reply
"Answer: insufficient evidence".
```

The assistant turn is prefilled with `Answer:`. HoVer and FEVER use the same template with *Claim* in place of *Question*, ask for one of the dataset's labels, and have no abstention line. The claim extractor of Experiment 0 (version `extract-v2`) uses the system message *You split sentences into short, simple facts.* and asks for one fact per line, naming the subject instead of using pronouns and using only information stated in the sentence. Its output is prefilled with `-` .

B Serialisation examples

The following renderings are produced by the experiment code for a MuSiQue development question that was used only during prompt development and is excluded from every reported run: *What character in Princess Diaries does the actress play who was also a cast member in On Golden Pond?* (gold: *Queen dowager Clarisse Renaldi*). Long sentences are abbreviated here with "…".

R0 (passages).

```
[E1] (source: The Princess Diaries
(film)) The Princess Diaries is a 2001
American teen romantic comedy ... It
stars Anne Hathaway ... ruled by her
grandmother Queen dowager Clarisse
Renaldi (Julie Andrews). ...
[E2] (source: On Golden Pond (2001
film)) On Golden Pond is a 2001
American television adaptation of the
play starring Julie Andrews and
Christopher Plummer. ...
```

R1 (sentences, dataset order).

```
[E1] (source: The Princess Diaries
(film)) The Princess Diaries is a 2001
American teen romantic comedy ...
[E2] (source: The Princess Diaries
(film)) It stars Anne Hathaway ...
Queen dowager Clarisse Renaldi (Julie
Andrews).
[E3] (source: The Princess Diaries
(film)) The film also stars ...
[E4] (source: On Golden Pond (2001
film)) On Golden Pond is a 2001
American television adaptation of the
play starring Julie Andrews and
Christopher Plummer.
[E5] ... [E6] ...
```

Item E2 begins with a pronoun. Separated from its passage, its subject is recoverable only through the source title that every item carries. We record such context-dependent facts per question; they are one way in which an information-equivalent transformation can still change what is easy to read.

R3 (steps), R5 (graph order) and R4 (graph order plus relations). R3 groups the same six items under `Step 1:` (the three *On Golden Pond* sentences) and `Step 2:` (the three *Princess Diaries* sentences), following the MuSiQue decomposition. R5 lists the same items in the same order without step headers. R4 is R5 followed by

```
Relations:
E5 DEPENDS_ON E1
```

i.e. the sentence naming the character depends on the sentence establishing that Julie Andrews was in *On Golden Pond*. This edge is derived from the MuSiQue decomposition with an exact anchor; it is not human-verified.

C Reconstruction of the EG-RAG baseline

B4 follows the stage order of the EG-RAG algorithm: hybrid retrieval, reranking, claim extraction, relation prediction, belief and utility propagation, conflict resolution, budgeted subgraph selection and serialisation. Every component that the original description leaves unspecified is our choice:

- BM25 plus bge-small-en-v1.5 with min–max normalisation and equal weights;
- ms-marco-MiniLM-L6-v2 reranking;
- the 7B generator as claim extractor, with a lexical-support filter;
- nli-deberta-v3-xsmall for support, contradiction and duplication, with thresholds 0.7 and 0.9;
- a title-mention heuristic for dependencies;
- years as the only temporal signal;
- our own feature weights, a 450-token budget and at most 12 claims;
- greedy connected selection.

The full ledger is in the repository (`docs/NOVELTY_AND_BASELINES.md`). B4 is therefore a reconstruction under our choices. Its results say nothing about the original implementation, and no result of the original paper is reused.

D Perturbations

Experiment 2 applies each intervention to the R1 and R4 renderings of the gold evidence with a recorded seed (0). The rates are $\{0.2, 0.4\}$ for P1–P4 and 0.4 for P5 and P6, each against the same question's rate-0 record, and $k = \lceil \text{rate} \cdot n_{\text{eligible}} \rceil$ items are affected.

- **P1** removes required (gold-support) facts.
- **P2** retypes or flips derived relations (R4 only).
- **P3** permutes item positions.
- **P4** inserts real non-gold sentences from the question's own candidate pool, chosen by BM25 similarity (*hard*), and never sentences containing a gold answer.
- **P5** strips source titles.
- **P6** deletes a negation, or replaces an entity in a fact with *it*.

Every changed item is recorded before and after. P1 and P6 can remove the answer and are therefore not label-preserving; we report their answer-change and abstention rates but no accuracy change. An intervention is unavailable when nothing is eligible:

- P2 on R1 or on questions without edges;
- P3 with a single item;
- P4 on HoVer and FEVER, which have no candidate pool;
- P6 when no fact contains a negation;
- P7 everywhere, since no verified temporal data exist.

These cases are counted and never scored as zero.

E Details of the exploratory pilot results

Evidence loss under a token budget (pilot). Protocol B was run only in the exploratory pilot (Figure 4 and Table 8, Appendix F). Its accounting of what each condition showed is deterministic and does not depend on the generator. Under a budget equal to the content tokens of the fact set, with each representation keeping as much of the shared ranking as it can afford, R0 keeps 0.83 (2WikiMultiHopQA) and 0.81 (MuSiQue) of the facts, whereas R1 keeps 0.64 and 0.65 and R4 keeps 0.63 and 0.62, because their identifiers, titles and relation lines cost 104–108 (2Wiki) and 81–88 (MuSiQue) tokens of overhead against 28 and 32 for R0. The verbose formats also lose the facts that matter: R0 drops 0.11 answer-bearing facts per question on 2WikiMultiHopQA against 0.33 for R1 and 0.34 for R4 (MuSiQue: 0.19, 0.28 and 0.35), so that the answer string remains in 0.85 of R0 blocks but only 0.66 of R1 and 0.64 of R4 blocks. The corresponding accuracy differences are smaller than these evidence losses and inconclusive at $n = 100$. For example, R1−R0 is -0.05 $[-0.11, +0.01]$ on 2WikiMultiHopQA, and R4−R5 is -0.05 $[-0.11, +0.01]$ on MuSiQue. Under the shared policy, every representation shows the same facts and pays the same evidence loss. Halving the budget reduces exact match for every representation to 0.08–0.19.

Retrieved-evidence baselines (pilot). Table 9 (Appendix F) reports Experiment 0 on the 200 primary pilot questions. Reranking (B1) improves on BM25 (B0) on MuSiQue by $+0.06$ $[0.00, +0.12]$, and the 2WikiMultiHopQA difference is inconclusive. Extracted claims (B2) and predicted reasoning paths (B3) are indistinguishable from reranked passages in exact match; the extracted claims keep the answer string as often as the passages they come from (0.78 and 0.44), but at roughly three times the evidence tokens (1,390–1,711 against 501–538), because every passage sentence becomes several facts. Our reconstruction of EG-RAG (B4) uses a similar number of tokens to B1 (442–489 against 501–538) but is less accurate, by -0.15 $[-0.23, -0.07]$ on 2WikiMultiHopQA and -0.09 $[-0.15, -0.04]$ on MuSiQue: its budgeted selection keeps the answer string less often (0.56 against 0.78; 0.31 against 0.44), and coverage falls by 0.29 and 0.15.

Controlled perturbations (pilot). Table 10 and Figure 3 (Appendix F) summarise Experiment 2. These comparisons are many (roughly fifty cells), and we expect a few intervals to exclude zero by chance; we report the patterns that recur across datasets or representations. When required facts are removed (P1), the model mostly abstains rather than guessing: abstention rises from 0.43 to 0.72 on MuSiQue (R1) and from 0.24 to 0.48 on 2WikiMultiHopQA. Hard distractors (P4) reduce exact match in every QA cell, by up to 0.16 (HotpotQA R1, $[-0.28, -0.04]$) and by 0.09 on MuSiQue R1 ($[-0.17, -0.01]$), and raise abstention on MuSiQue. Retyping or flipping 40% of the derived relations (P2, R4 only) lowers 2WikiMultiHopQA exact match by 0.10 $[-0.18, -0.01]$ on the 71 questions that have relations, and is inconclusive on MuSiQue and HotpotQA. Reordering (P3) and stripping source titles (P5) produce small effects of inconsistent sign, and HoVer and FEVER verdicts are almost unaffected by any intervention.

On the secondary datasets (50 questions each; run `C_exp0`), every Experiment 0 contrast is inconclusive, e.g. B4−B1 is 0.00 $[-0.12, +0.12]$ EM on HotpotQA and -0.06 $[-0.22, +0.12]$ verdict accuracy on HoVer.

F Additional results

Table 7 gives the absolute scores of every representation under Protocol A in the expanded pilot, Table 8 the full information accounting under Protocols A and B, Table 9 the retrieved-evidence baselines, and Table 10 and Figure 3 the perturbation experiment. Table 4 lists the secondary contrasts on the confirmatory sample and Table 6 the analysis of predicted contradiction lines; the remaining tables and figures are exploratory pilot results.

Data	Contrast	n	ΔEM	95% CI	Ident.	ΔCov.
MuSiQue	R2−R1	1744	+0.013	[-0.001, +0.028]	2	+0.041
MuSiQue	R3−R5	1744	-0.007	[-0.016, +0.002]	0	-0.015
MuSiQue	R4P−R4	1744	-0.045	[-0.060, -0.030]	94	-0.089
2Wiki	R2−R1	3232	-0.024	[-0.036, -0.012]	4	-0.017
2Wiki	R3−R5	3232	-0.004	[-0.016, +0.008]	0	-0.015
2Wiki	R4P−R4	3232	-0.036	[-0.045, -0.027]	818	-0.061

Table 4: Secondary contrasts on the confirmatory sample (pre-specified as secondary; not part of the tested family, so descriptive). Source: `F_exp1_contrasts.csv`.

Data	Contrast	ΔEM	both ans.	one abstains	abst. only first / only second	neither
2Wiki	R1−R0	-0.017	+0.001	-0.018	250/81	529
2Wiki	R4−R5	+0.006	+0.010	-0.004	123/88	731
2Wiki	R4P−R4	-0.036	-0.011	-0.025	257/59	795
2Wiki	R5−R1	-0.008	-0.004	-0.004	170/130	649
MuSiQue	R1−R0	-0.001	-0.002	+0.001	94/110	710
MuSiQue	R4−R5	+0.010	+0.006	+0.004	82/89	709
MuSiQue	R4P−R4	-0.045	-0.005	-0.040	218/63	728
MuSiQue	R5−R1	-0.011	-0.012	+0.001	98/104	700

Table 5: Exploratory decomposition of each paired EM difference on the confirmatory sample (added after the confirmatory result). *both ans.*: contribution of questions answered under both representations; *one abstains*: contribution of questions answered under only one. The columns sum to ΔEM. Source: `F_exp1_decomposition_exploratory.csv`.

Data	R4P shows CONTRADICTS	n	Abst. R4	Abst. R4P	EM R4	EM R4P
2Wiki	yes	2034	0.32	0.41	0.33	0.28
2Wiki	no	1198	0.17	0.17	0.48	0.48
MuSiQue	yes	1010	0.50	0.63	0.25	0.18
MuSiQue	no	734	0.39	0.42	0.33	0.32

Table 6: Exploratory: the same questions under derived relations (R4) and predicted relations (R4P), split by whether the predicted block contains at least one `CONTRADICTS` line (an input property). Source: `F_exp1_r4p_relations_exploratory_paired.csv`.

Data	R0	R1	R2	R3	R4	R4P	R5
MuSiQue	0.28 (0.57)	0.31 (0.57)	0.28 (0.59)	0.24 (0.52)	0.29 (0.59)	0.21 (0.42)	0.29 (0.58)
2Wiki	0.44 (0.84)	0.37 (0.76)	0.36 (0.74)	0.37 (0.67)	0.41 (0.69)	0.38 (0.66)	0.38 (0.70)
HotpotQA	0.40 (0.88)	0.42 (0.82)	0.35 (0.85)	0.40 (0.86)	0.40 (0.84)	0.42 (0.80)	0.38 (0.82)
HoVer	0.68 (1.00)	0.68 (1.00)	0.73 (1.00)	–	0.68 (1.00)	0.74 (1.00)	0.68 (1.00)
FEVER	0.96 (1.00)	0.96 (1.00)	0.97 (1.00)	–	0.96 (1.00)	0.96 (1.00)	0.96 (1.00)

Table 7: Exploratory pilot, Protocol A: EM (QA) or verdict accuracy (HoVer, FEVER), with coverage (answered rate) in parentheses, for the same facts in seven representations. n=100 (MuSiQue, 2Wiki) or 50. R3 is unavailable without a reasoning order. Source: `C_exp1_absolute.csv`.

Data	Setting	Rep.	Kept	Ovh.	Tok.	Ans. drop	Ans. in ev.	EM	Cov.
MuSiQue	A	R0	1.00	33	378	0.00	0.99	0.28	0.57
MuSiQue	A	R1	1.00	122	467	0.00	0.99	0.31	0.57
MuSiQue	A	R5	1.00	122	467	0.00	0.99	0.29	0.58
MuSiQue	A	R4	1.00	141	486	0.00	0.99	0.29	0.59
MuSiQue	B 1.0× per-rep.	R0	0.81	32	327	0.19	0.89	0.25	0.52
MuSiQue	B 1.0× per-rep.	R1	0.65	81	322	0.28	0.85	0.26	0.52
MuSiQue	B 1.0× per-rep.	R5	0.65	81	322	0.28	0.85	0.22	0.53
MuSiQue	B 1.0× per-rep.	R4	0.62	88	321	0.35	0.80	0.17	0.44
MuSiQue	B 1.0× shared	R0	0.60	30	251	0.38	0.77	0.18	0.47
MuSiQue	B 1.0× shared	R1	0.60	75	296	0.38	0.77	0.20	0.48
MuSiQue	B 1.0× shared	R5	0.60	75	296	0.38	0.77	0.18	0.52
MuSiQue	B 1.0× shared	R4	0.60	86	307	0.38	0.77	0.17	0.46
MuSiQue	B 0.5× per-rep.	R0	0.34	23	144	0.71	0.49	0.10	0.30
MuSiQue	B 0.5× per-rep.	R1	0.30	38	143	0.77	0.43	0.09	0.30
MuSiQue	B 0.5× per-rep.	R5	0.30	38	143	0.77	0.43	0.09	0.30
MuSiQue	B 0.5× per-rep.	R4	0.29	39	142	0.79	0.42	0.08	0.31
MuSiQue	B 0.5× shared	R0	0.27	21	120	0.80	0.42	0.09	0.28
MuSiQue	B 0.5× shared	R1	0.27	35	133	0.80	0.42	0.09	0.30
MuSiQue	B 0.5× shared	R5	0.27	35	133	0.80	0.42	0.09	0.30
MuSiQue	B 0.5× shared	R4	0.27	38	136	0.80	0.42	0.08	0.30
2Wiki	A	R0	1.00	29	414	0.00	0.98	0.44	0.84
2Wiki	A	R1	1.00	157	543	0.00	0.98	0.37	0.76
2Wiki	A	R5	1.00	157	543	0.00	0.98	0.38	0.70
2Wiki	A	R4	1.00	167	552	0.00	0.98	0.41	0.69
2Wiki	B 1.0× per-rep.	R0	0.83	28	366	0.11	0.85	0.38	0.78
2Wiki	B 1.0× per-rep.	R1	0.64	104	361	0.33	0.66	0.33	0.67
2Wiki	B 1.0× per-rep.	R5	0.64	104	361	0.33	0.66	0.31	0.64
2Wiki	B 1.0× per-rep.	R4	0.63	108	361	0.34	0.64	0.33	0.64
2Wiki	B 1.0× shared	R0	0.60	26	265	0.39	0.60	0.30	0.67
2Wiki	B 1.0× shared	R1	0.60	97	337	0.39	0.60	0.27	0.63
2Wiki	B 1.0× shared	R5	0.60	97	337	0.39	0.60	0.27	0.62
2Wiki	B 1.0× shared	R4	0.60	103	342	0.39	0.60	0.29	0.62
2Wiki	B 0.5× per-rep.	R0	0.39	21	173	0.59	0.43	0.17	0.50
2Wiki	B 0.5× per-rep.	R1	0.32	52	171	0.62	0.39	0.17	0.39
2Wiki	B 0.5× per-rep.	R5	0.32	52	171	0.62	0.39	0.19	0.41
2Wiki	B 0.5× per-rep.	R4	0.31	52	170	0.62	0.39	0.15	0.38
2Wiki	B 0.5× shared	R0	0.29	20	131	0.63	0.38	0.14	0.45
2Wiki	B 0.5× shared	R1	0.29	48	160	0.63	0.38	0.15	0.36
2Wiki	B 0.5× shared	R5	0.29	48	160	0.63	0.38	0.16	0.37
2Wiki	B 0.5× shared	R4	0.29	50	161	0.63	0.38	0.15	0.37

Table 8: Information accounting under Protocol A and Protocol B (budget $\beta\times$ the content tokens of the fact set). Kept: fraction of facts shown; Ovh./Tok.: mean overhead and evidence tokens; Ans. drop: answer-bearing facts cut per question; Ans. in ev.: answer string present in the block. Source: `C_exp1_protocol_accounting.csv`, `C_exp1_absolute.csv`.

Data	Sys.	EM	95% CI	F1	Cov.	Tok.	Units	Ans. in ev.	Lat. (s)
MuSiQue	B0	0.06	[+0.02, +0.11]	0.10	0.23	493	4.0	0.34	0.36
MuSiQue	B1	0.12	[+0.06, +0.19]	0.17	0.33	538	4.0	0.44	0.36
MuSiQue	B2	0.13	[+0.07, +0.20]	0.17	0.36	1390	57.1	0.44	0.86
MuSiQue	B3	0.15	[+0.08, +0.22]	0.18	0.38	1667	57.1	0.44	0.91
MuSiQue	B4	0.03	[+0.00, +0.07]	0.05	0.18	442	11.9	0.31	0.51
2Wiki	B0	0.23	[+0.16, +0.31]	0.27	0.54	506	4.0	0.69	0.38
2Wiki	B1	0.26	[+0.18, +0.35]	0.30	0.54	501	4.0	0.78	0.39
2Wiki	B2	0.24	[+0.16, +0.33]	0.27	0.48	1436	56.4	0.78	0.67
2Wiki	B3	0.20	[+0.12, +0.28]	0.24	0.50	1711	56.4	0.78	0.68
2Wiki	B4	0.11	[+0.05, +0.17]	0.13	0.25	489	11.9	0.56	0.51

Table 9: Experiment 0, retrieved evidence (exploratory, n=100 per dataset). Tok.: mean evidence tokens; Units: items shown; Lat.: answer-generation latency (claim extraction for B2–B4 is not included). B4 is our reconstruction of EG-RAG. Source: `A_exp0_absolute.csv`.

Data	Rep.	Perturbation	n	Lab.-pres.	ΔEM	95% CI	Ans. changed	Abstention
MuSiQue	R1	P1	100	0	–	–	0.40	0.43→0.72
MuSiQue	R1	P3	100	100	-0.07	[-0.13, -0.02]	0.13	0.43→0.44
MuSiQue	R1	P4 hard	100	100	-0.09	[-0.17, -0.01]	0.27	0.43→0.51
MuSiQue	R1	P5 strip	100	100	-0.03	[-0.08, +0.02]	0.12	0.43→0.41
MuSiQue	R1	P6 entity	100	0	–	–	0.14	0.43→0.50
MuSiQue	R1	P6 negation	29	0	–	–	0.03	0.52→0.55
MuSiQue	R4	P1	100	0	–	–	0.44	0.41→0.71
MuSiQue	R4	P2	100	100	-0.01	[-0.07, +0.05]	0.24	0.41→0.47
MuSiQue	R4	P3	100	100	-0.03	[-0.08, +0.02]	0.14	0.41→0.44
MuSiQue	R4	P4 hard	100	100	-0.08	[-0.16, +0.00]	0.34	0.41→0.54
MuSiQue	R4	P5 strip	100	100	+0.00	[-0.04, +0.04]	0.10	0.41→0.38
MuSiQue	R4	P6 entity	100	0	–	–	0.19	0.41→0.43
MuSiQue	R4	P6 negation	29	0	–	–	0.03	0.48→0.52
2Wiki	R1	P1	100	0	–	–	0.43	0.24→0.48
2Wiki	R1	P3	100	100	+0.04	[-0.04, +0.12]	0.30	0.24→0.19
2Wiki	R1	P4 hard	100	100	-0.04	[-0.13, +0.05]	0.35	0.24→0.31
2Wiki	R1	P5 strip	100	100	+0.02	[-0.03, +0.08]	0.16	0.24→0.18
2Wiki	R1	P6 entity	100	0	–	–	0.14	0.24→0.27
2Wiki	R1	P6 negation	15	0	–	–	0.00	0.20→0.20
2Wiki	R4	P1	100	0	–	–	0.43	0.31→0.51
2Wiki	R4	P2	71	71	-0.10	[-0.18, -0.01]	0.44	0.35→0.34
2Wiki	R4	P3	100	100	-0.01	[-0.09, +0.07]	0.26	0.31→0.26
2Wiki	R4	P4 hard	100	100	-0.03	[-0.12, +0.06]	0.35	0.31→0.36
2Wiki	R4	P5 strip	100	100	+0.08	[+0.01, +0.16]	0.19	0.31→0.26
2Wiki	R4	P6 entity	100	0	–	–	0.17	0.31→0.33
2Wiki	R4	P6 negation	15	0	–	–	0.00	0.40→0.40
HotpotQA	R1	P1	50	0	–	–	0.36	0.18→0.44
HotpotQA	R1	P3	50	50	-0.06	[-0.16, +0.04]	0.18	0.18→0.14
HotpotQA	R1	P4 hard	50	50	-0.16	[-0.28, -0.04]	0.26	0.18→0.24
HotpotQA	R1	P5 strip	50	50	-0.02	[-0.08, +0.04]	0.08	0.18→0.14
HotpotQA	R1	P6 entity	50	0	–	–	0.18	0.18→0.18
HotpotQA	R1	P6 negation	5	0	–	–	0.00	0.20→0.20
HotpotQA	R4	P1	50	0	–	–	0.42	0.16→0.44
HotpotQA	R4	P2	25	25	-0.04	[-0.24, +0.12]	0.28	0.08→0.16
HotpotQA	R4	P3	50	50	-0.04	[-0.14, +0.06]	0.14	0.16→0.16
HotpotQA	R4	P4 hard	50	50	-0.10	[-0.24, +0.04]	0.34	0.16→0.20
HotpotQA	R4	P5 strip	50	50	+0.02	[-0.04, +0.08]	0.08	0.16→0.14
HotpotQA	R4	P6 entity	50	0	–	–	0.16	0.16→0.18
HotpotQA	R4	P6 negation	5	0	–	–	0.00	0.20→0.20

Table 10: Experiment 2 at rate 0.4 (exploratory). ΔEM is computed only on label-preserving records without introduced answers (Lab.-pres.); P1 and P6 can remove the answer, so only answer-change and abstention rates are given. Source: `C_exp2_sensitivity.csv`.

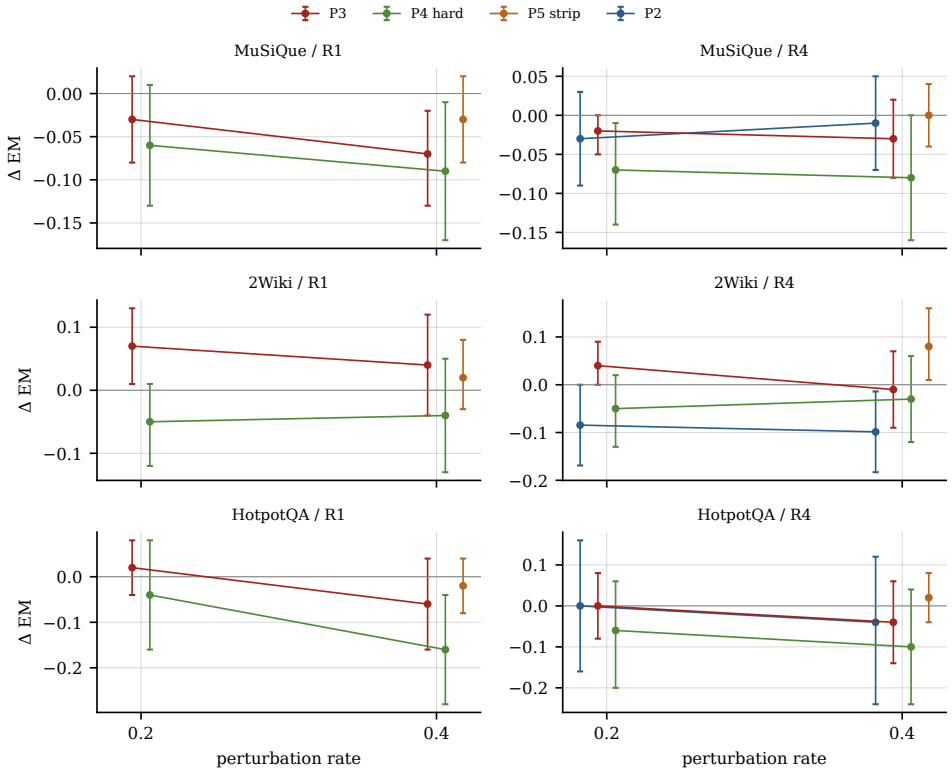


Figure 3: Exploratory pilot: change in exact match under label-preserving perturbations, relative to the same question unperturbed (95% CIs). P1 and P6 can remove the answer and are reported in Table 10 only.

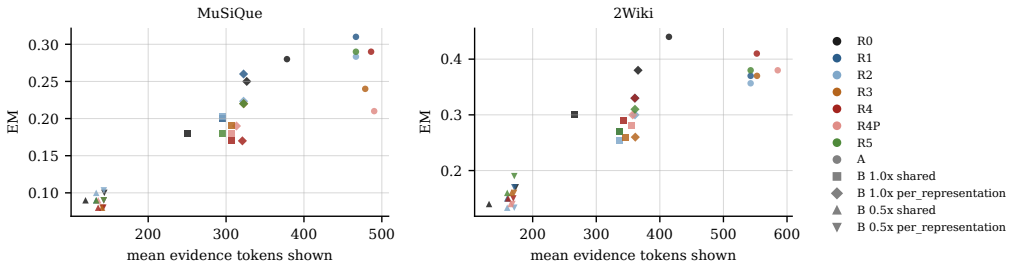


Figure 4: Exploratory pilot: exact match against the mean number of evidence tokens shown, for Protocol A (natural length) and Protocol B (budget $1.0\times$ or $0.5\times$ the content tokens; shared or per-representation policy).